

YUNCONG YANG

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EDUCATION

University of Massachusetts, Amherst

Ph.D. student in Computer Science

Advisor: Chuang Gan

Research Interests: Embodied AI, Multi-modal Foundation Models, Robotics

Amherst, MA

Sep 2024 - Present

Columbia University

M.S. in Computer Science

Advisor: Shih-Fu Chang

Research Interests: Computer Vision, Multi-modal Foundation Models

New York, NY

Graduated Dec 2022

Columbia University

B.S. in Computer Science: Intelligent Systems Track

GPA: 4.05/4.33 (Top 5%) | Summa Cum Laude

New York, NY

Graduated May 2021

PUBLICATIONS

3D-Mem: 3D Scene Memory for Embodied Exploration and Reasoning

CVPR 2025

Yuncong Yang*, Han Yang*, Jiachen Zhou, Peihao Chen, Hongxin Zhang, Yilun Du, Chuang Gan

TempCLR: Temporal Alignment Representation with Contrastive Learning

ICLR 2023

Yuncong Yang*, Jiawei Ma*, Shiyuan Huang, Long Chen, Xudong Lin, Guangxing Han, Shih-Fu Chang

MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge

NeurIPS 2022

[Outstanding Paper Award, Featured Paper Presentation]

Linxi “Jim” Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, **Yuncong Yang**, Haoyi Zhu, Andrew Tang, De-An Huang, Yuke Zhu[†], Anima Anandkumar[†]

Few-Shot End-to-End Object Detection via Constantly Concentrated Encoding Across Heads

ECCV 2022

Jiawei Ma, Guangxing Han, Shiyuan Huang, **Yuncong Yang**, Shih-Fu Chang

RESEARCH EXPERIENCE

University of Massachusetts, Amherst

Doctoral Research Assistant | Advisor: Prof. Chuang Gan

Sep 2024 - Now

 *Embodied AI and 3D Scene Understanding*

- Developed 3D-Mem, a 3D scene memory framework that empowers VLM agents with lifelong exploration and reasoning abilities in 3D environments.

MIT-IBM Watson AI Lab

Research Assistant Intern | Advisor: Prof. Chuang Gan

Apr 2023 - Aug 2024

 *Foundation Models for Robotics*

- To address underfitting issues caused by heterogeneous datasets and varying observation spaces in large-scale robotics training, we integrated a Mixture of Experts (MoE) architecture into the Decision Transformer.

Nvidia

Research Assistant Intern | Advisor: Dr. Jim Fan

Feb 2022 - Feb 2023

🏠 Embodied AI and LLM Agents

- Contributed to MineDojo, a Minecraft-based benchmark for open-ended embodied agents, by leading the design of its comprehensive task suite and assisting in the development of MineCLIP, a video-language model that serves as a reward function for agents.
- Contributed to Voyager, an LLM-powered code generation agent on Minecraft, by designing Python APIs for MineDojo to support efficient code generation, comprehensive data generation, and training pipelines.

Columbia University

Research Assistant | Advisor: Prof. Shih-Fu Chang

Dec 2021 - Dec 2022

🎬 Video Understanding

- Developed a sequence-level contrastive learning framework that leverages temporal order consistency for long-video understanding, achieving significant performance improvements in video-text retrieval, action step localization, and few-shot video action recognition.

PROJECTS

Dynamic Grasping with Moving Obstacles

Advisor: Prof. Shuran Song

Sep 2021 - Dec 2021

- Tackled the problem of grasping a moving item with moving obstacles by modeling the target item and obstacles as 3D bounding boxes for real-time prediction of their future poses.

Adversarial Training for Few-Shot Classification

Advisor: Prof. Junfeng Yang

Sep 2021 - Dec 2021

- Enhanced the effectiveness of adversarial training for few-shot classifier by attacking query-set and support-set simultaneously.

AWARDS & HONORS

- Outstanding Paper Award, *NeurIPS 2022 Datasets and Benchmarks Track*
- Summa Cum Laude, *Columbia University*
- Student Member, *Phi Beta Kappa*
- Dean's List (8 semesters)

TEACHING EXPERIENCES

Introduction to Computation, University of Massachusetts, Amherst

Fall 2024

Computer Vision II, Columbia University

Spring 2022